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United States Patent Application Publication

20250279368

Kind Code

A1

Publication Date

September 04, 2025

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REDISTRIBUTION LAYERS WITH CARBON-BASED CONDUCTIVE ELEMENTS, AND RELATED MICROELECTRONIC DEVICE PACKAGES AND METHODS OF FABRICATION

Abstract

Semiconductor device packages include a redistribution layer (RDL) with carbon-based conductive elements. The carbon-based material of the RDL may have low electrical resistivity and may be thin (eg., less than about 0.2 μm). Adjacent passivation material may also be thin (eg., less than about 0.2 μm). Methods for forming the semiconductor device packages include forming the carbon-based material (eg., at high temperatures (e.g., at least about 550° C.)) on an initial support wafer with a sacrificial substrate. Later or separately, components of a device region of the package may be formed and then joined to the initial support wafer before the sacrificial substrate is removed to leave the carbon-based material joined to the device region.

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Family ID: 71124140

Appl. No.: 19/203876

Filed: May 09, 2025

Related U.S. Application Data

parent US continuation 18460481 20230901 parent-grant-document US 12300622 child US 19203876

parent US division 17076602 20201021 parent-grant-document US 11749608 child US 18460481

parent US division 16236681 20181231 parent-grant-document US 10854549 child US 17076602

Publication Classification

Int. Cl.: H01L23/538 (20060101); **H01L23/532** (20060101); **H01L25/11** (20060101)

U.S. Cl.:

CPC H01L23/5384 (20130101); **H01L23/53233** (20130101); **H01L23/53261** (20130101); **H01L23/53276** (20130101); **H01L23/5385** (20130101); **H01L23/5386** (20130101); **H01L25/117** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/460,481, filed Sep. 1, 2023, which is a divisional of U.S. patent application Ser. No. 17/076,602, filed Oct. 21, 2020, now U.S. Pat. No. 11,749,608, issued Sep. 5, 2023, which is a divisional of U.S. patent application Ser. No. 16/236,681, filed Dec. 31, 2018, now U.S. Pat. No. 10,854,549, issued Dec. 1, 2020, the disclosure of each of which is hereby incorporated in its entirety herein by this reference.

TECHNICAL FIELD

[0002] The disclosure, in various embodiments, relates generally to redistribution layers, methods of fabrication, and semiconductor device packaging. More particularly, embodiments disclosed herein relate to redistribution layers comprising carbon-based conductive elements, methods for fabricating same, and related semiconductor device packages and systems.

BACKGROUND

[0003] As the electronics industry evolves and encompasses more and different applications (e.g., smart phones and other mobile devices, increasingly compact personal computers (e.g., laptops, tablets), artificial intelligence, the Internet of Things, and cloud computing), there is an ever-increasing demand for high density, small form factor modular packages comprising multiple semiconductor devices, such as in packages having stacked semiconductor dice. In such structures, form factor may be a significant consideration in designing mobile applications, among others, wherein significant form factor considerations include not only the footprint (e.g., length and width) of a package but also the height of the package.

[0004] Packages may include a redistribution layer (RDL) in which the contacts of semiconductor die are redistributed over a larger area. The RDL is typically defined by the addition and patterning of metal materials (e.g., in the form of conductive traces and/or vias) and dielectric materials (e.g., passivation materials) onto the surface of a substrate to re-route an Input/Output (I/O) layout into a looser pitch footprint. Such redistribution may typically use metal materials (e.g., aluminum or copper) to reroute peripheral pads to an area array configuration. To achieve effective electrical conductivity and to compensate for resistivity of the metal materials, the metal materials of the conductive traces and vias are often several micrometers in thickness, as are the adjacent dielectric materials. Designing and fabricating the RDL of a package with small height form factors that still effectively conduct electricity for redistribution continues to present challenges.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a cross-sectional, elevational, schematic illustration of a semiconductor device package according to an embodiment of the disclosure.

[0006] FIG. 2 is an enlarged and more detailed schematic illustration of a portion of the semiconductor device package of FIG. 1.

[0007] FIG. 3 is a cross-sectional, elevational, schematic illustration of a singulated semiconductor device package according to an embodiment of the disclosure.

[0008] FIGS. 4 through 15 are cross-sectional, elevational, schematic illustrations during various stages of processing to fabricate the semiconductor device package of FIGS. 1 and 2, according to embodiments of the disclosure, wherein the stages of the illustrations of FIGS. 6 through 8 follow that of FIG. 4 and precede that of FIG. 5 according to an embodiment of the disclosure, and wherein the stages of the illustrations of FIGS. 9 through 11 follow that of FIG. 4 and precede that of FIG. 5 according to another embodiment of the disclosure.

[0009] FIGS. 16 through 20, in conjunction with FIGS. 4 and 6, are cross-sectional, elevational, schematic illustrations during various stages of processing to fabricate a semiconductor device package according to an embodiment of the disclosure, wherein FIG. 16 illustrates a stage following that of FIG. 6.

[0010] FIGS. 21 and 22, in conjunction with FIGS. 4 through 15, are cross-sectional, elevational, schematic illustrations during various stages of processing to fabricate a semiconductor device package according to an embodiment of the disclosure, wherein FIG. 21 illustrates a stage following that of FIG. 15.

[0011] FIGS. 23 through 25, in conjunction with FIGS. 4, 6 and 16 through 18, are cross-sectional, elevational, schematic illustrations during various stages of processing to fabricate a semiconductor device package according to an embodiment of the disclosure, wherein FIG. 23 illustrates a stage following that of FIG. 18.

[0012] FIGS. 26 through 31 are cross-sectional, elevational, schematic illustrations during various stages of processing to fabricate a semiconductor device package according to an embodiment of the disclosure.

[0013] FIGS. 32 through 36 are cross-sectional, elevational, schematic illustrations during various stages of processing to fabricate a semiconductor device package according to an embodiment of the disclosure.

[0014] FIGS. 37 through 41 are cross-sectional, elevational, schematic illustrations during various stages of processing to fabricate the singulated semiconductor device package of FIG. 3.

[0015] FIG. 42 is a block diagram of an electronic system including at least one semiconductor device package according to one or more embodiments of the disclosure.

DETAILED DESCRIPTION

[0016] Semiconductor device packages and methods of fabrication therefore are disclosed in which the packages include a redistribution layer (RDL) comprising carbon-based conductors (eg., carbon-based conductive elements). The carbon-based material of the redistribution layer may exhibit extremely low electrical resistivity, enabling the redistribution layer to effectively transmit power, ground, and data signals at fast rates even when only a thin (eg., of a relatively small vertical height) region of the carbon-based material is present. The thinness of the redistribution layer may also enable surrounding insulative regions (eg., regions of interdielectric material, regions of passivation material) to be thin and still function as insulators. That is, with a thin redistribution layer, good coverage (eg., good “step coverage” (eg., uniform coverage)) of the material of the redistribution layer by surrounding insulative material may be more reliably achieved with a lesser amount (eg., thickness) of the insulative material than needed for a thicker redistribution layer. For example, lower thicknesses of the redistribution layer may enable a lesser amount of a passivation material to ensure adequate coverage of the sidewalls of the structures of the redistribution layer. Thus, a thinner redistribution layer may enable a thinner surrounding insulative region to achieve sufficient coverage and insulation. Also, the carbon-based material may be formed, with a high temperature formation process, on a substrate separate from a wafer on which other semiconductor device components are formed. Thus, the high temperature process for forming the carbon-based material may not detrimentally impact the fabrication of the other semiconductor device components. The resulting semiconductor device packages exhibit a very

low height form factor, enabling improved scaling in three-dimensional stacks that include such packages.

[0017] As used herein, the term “redistribution layer” means and includes a region comprising conductive elements and placed on a surface of a semiconductor die bearing bond pads or other terminals, which region enables component-to-component electrical connections at select, laterally offset locations of the semiconductor die and another component. A redistribution layer (RDL) may comprise at least one conductive region (e.g., a single layer of conductive traces, or multiple layers interconnected vertically) and at least one region of insulative (e.g., passivation) material adjacent the conductive material.

[0018] As used herein, the term “carbon-based,” when referring to a material, means and includes a material comprising at least 50 at. % carbon, consisting essentially of carbon or a carbon allotrope, or consisting of carbon or a carbon allotrope. “Carbon-based” materials include, but are not necessarily limited to, the carbon allotropes of graphene, carbon nanotubes, carbon nanobuds, and carbon nanoribbons.

[0019] As used herein, the term “low resistivity,” when referring to an exhibited property of a material, refers to an electrical resistance or electrical resistivity on the order of about 10^{-6} $\Omega \cdot \text{cm}$ or less.

[0020] As used herein, the term “substrate” means and includes a base material or other construction upon which components, such as those within memory cells, are formed. The substrate may be a semiconductor substrate, a base semiconductor material on a supporting structure, a metal electrode, or a semiconductor substrate having one or more materials, structures, or regions formed thereon. The substrate may be a conventional silicon substrate or other bulk substrate including a semiconductive material. As used herein, the term “bulk substrate” means and includes not only silicon wafers, but also silicon-on-insulator (“SOI”) substrates, such as silicon-on-sapphire (“SOS”) substrates or silicon-on-glass (“SOG”) substrates, epitaxial layers of silicon on a base semiconductor foundation, or other semiconductor or optoelectronic materials, such as silicon-germanium ($\text{Si}_{1-x}\text{Ge}_x$, where x is, for example, a mole fraction between 0.2 and 0.8), germanium (Ge), gallium arsenide (GaAs), gallium nitride (GaN), or indium phosphide (InP), among others. Furthermore, when reference is made to a “substrate” in the following description, previous process stages may have been utilized to form materials, regions, or junctions in the base semiconductor structure or foundation.

[0021] As used herein, the term “wafer” means and includes not only a conventional, substantially circular wafer of a single material (e.g., a single crystal silicon), but also other bulk substrates in the form of a silicon-on-insulator (SOI) structure, as well as bulk substrates comprising other materials (e.g., semiconductor materials (e.g., germanium, gallium arsenide)), as well as substrates employed in fabrication of structures thereon, such as RDLs, which substrates may comprise other materials such as glasses or ceramics, in addition to the foregoing.

[0022] As used herein, the terms “layer” and “film” mean and include a level, sheet, or coating of material residing on a structure, which level, sheet, or coating may be continuous or discontinuous between portions of the material, may be planar or non-planar, and may be conformal or non-conformal, unless otherwise indicated.

[0023] As used herein, the term “sacrificial,” when referring to a material, region, structure, or component, means and includes the material, region, structure, or component being of a preliminary or intermediary stage in a fabrication process and intended for removal, in whole or in substantial part, prior to completion of the fabrication process.

[0024] As used herein, the terms “lateral” or “horizontal” mean and include a direction that is parallel to a primary surface of the substrate on which the referenced material or structure is located. The width and length of a respective region or material may be defined as dimensions in a horizontal plane.

[0025] As used herein, the terms “longitudinal” or “vertical” mean and include a direction that is

perpendicular to a primary surface of the substrate on which a referenced material or structure is located. The height of a respective region or material may be defined as a dimension in a vertical plane.

[0026] As used herein, the terms “thickness,” “thinness,” or “height” mean and include a dimension in a straight-line direction that is normal to the closest surface of an immediately adjacent material or region of different composition.

[0027] As used herein, the term “between” is a spatially relative term used to describe the relative disposition of one material, region, or sub-region relative to at least two other materials, regions, or sub-regions. The term “between” may encompass both a disposition of one material, region, or sub-region directly adjacent to the other materials, regions, or sub-regions and a disposition of one material, region, or sub-region indirectly adjacent to the other materials, regions, or sub-regions.

[0028] As used herein, the term “proximate” is a spatially relative term used to describe disposition of one material, region, or sub-region near to another material, region, or sub-region. The term “proximate” includes dispositions of indirectly adjacent to, directly adjacent to, and internal to.

[0029] As used herein, the term “neighboring,” when referring to a material or region, means and refers to a next, most proximate material or region of an identified composition. Materials or regions of other compositions than the identified composition may be disposed between one material or region and its “neighboring” material or region of the identified composition. For example, a region of passivation material “neighboring” a region of conductive material is the passivation material region, eg., of a plurality of passivation material regions, that is next most proximate to the particular region of conductive material. The “neighboring” material or region may be directly or indirectly proximate the region or material of the identified composition.

[0030] As used herein, the terms “about” and “approximately,” when either is used in reference to a numerical value for a particular parameter, are inclusive of the numerical value and a degree of variance from the numerical value that one of ordinary skill in the art would understand is within acceptable tolerances for the particular parameter. For example, “about” or “approximately,” in reference to a numerical value, may include additional numerical values within a range of from 90.0 percent to 110.0 percent of the numerical value, such as within a range of from 95.0 percent to 105.0 percent of the numerical value, within a range of from 97.5 percent to 102.5 percent of the numerical value, within a range of from 99.0 percent to 101.0 percent of the numerical value, within a range of from 99.5 percent to 100.5 percent of the numerical value, or within a range of from 99.9 percent to 100.1 percent of the numerical value.

[0031] As used herein, the term “substantially,” when referring to a parameter, property, or condition, means and includes the parameter, property, or condition being equal to or within a degree of variance from a given value such that one of ordinary skill in the art would understand such given value to be acceptably met, such as within acceptable manufacturing tolerances. By way of example, depending on the particular parameter, property, or condition that is substantially met, the parameter, property, or condition may be “substantially” a given value when the value is at least 90.0% met, at least 95.0% met, at least 99.0% met, or even at least 99.9% met.

[0032] As used herein, reference to an element as being “on” or “over” another element means and includes the element being directly on top of, adjacent to (eg., laterally adjacent to, vertically adjacent to), underneath, or in direct contact with the other element. It also includes the element being indirectly on top of, adjacent to (eg., laterally adjacent to, vertically adjacent to), underneath, or near the other element, with other elements present therebetween. In contrast, when an element is referred to as being “directly on” or “directly adjacent to” another element, there are no intervening elements present.

[0033] As used herein, other spatially relative terms, such as “below,” “lower,” “bottom,” “above,” “upper,” “top,” and the like, may be used for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Unless otherwise specified, the spatially relative terms are intended to encompass different orientations of

the materials in addition to the orientation as depicted in the figures. For example, if materials in the figures are inverted, elements described as “below” or “under” or “on bottom of” other elements or features would then be oriented “above” or “on top of” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below, depending on the context in which the term is used, which will be evident to one of ordinary skill in the art. The materials may be otherwise oriented (rotated ninety degrees, inverted, etc.) and the spatially relative descriptors used herein interpreted accordingly.

[0034] As used herein, the terms “comprises,” “comprising,” “includes,” and/or “including” specify the presence of stated features, regions, stages, operations, elements, materials, components, and/or groups, but do not preclude the presence or addition of one or more other features, regions, stages, operations, elements, materials, components, and/or groups thereof.

[0035] As used herein, “and/or” includes any and all combinations of one or more of the associated listed items.

[0036] As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0037] As used herein, the terms “configured” and “configuration” mean and refer to a size, shape, material composition, orientation, and arrangement of a referenced material, structure, assembly, or apparatus so as to facilitate a referenced operation or property of the referenced material, structure, assembly or apparatus in a predetermined way.

[0038] The illustrations presented herein are not meant to be actual views of any particular component, device, or system, but are merely idealized representations that are employed to describe embodiments of the disclosure.

[0039] The following description provides specific details—such as material types, material thicknesses, and processing conditions—in order to provide a thorough description of embodiments of the disclosed packages, methods, and systems. However, a person of ordinary skill in the art will understand that the embodiments of the packages, methods, and systems may be practiced without employing these specific details. Indeed, the embodiments of the packages, methods, and systems may be practiced in conjunction with conventional semiconductor fabrication techniques employed in the industry.

[0040] Unless the context indicates otherwise, the materials described herein may be formed by any suitable technique including, but not limited to, spin coating, blanket coating, chemical vapor deposition (“CVD”), atomic layer deposition (“ALD”), plasma enhanced ALD, physical vapor deposition (“PVD”) (e.g., sputtering), or epitaxial growth. Depending on the specific material to be formed, the technique for depositing or growing the material may be selected by a person of ordinary skill in the art.

[0041] Unless the context indicates otherwise, the removal of materials described herein may be accomplished by any suitable technique including, but not limited to, etching (e.g., dry etching, wet etching, vapor etching), ion milling, abrasive planarization (eg., grinding or chemical mechanical planarization (CMP)), or other known methods.

[0042] The fabrication processes described herein do not form a complete process flow for processing semiconductor device structures. The remainder of the process flow, whether stages preceding or following those described, is known to those of ordinary skill in the art. Accordingly, only the methods and semiconductor device structures necessary to understand embodiments of the present structures, assemblies, systems, and methods are described herein.

[0043] Reference will now be made to the drawings, where like numerals refer to like components throughout. The drawings are not necessarily drawn to scale.

[0044] FIG. 1 illustrates a semiconductor device package **100** according to an embodiment of the disclosure. The semiconductor device package **100** includes a redistribution layer **102** (RDL) that includes a conductive carbon-based material **104** having low resistivity. Optionally, another conductive material (a “seed material” **106**) may be above the carbon-based material **104**.

[0045] In some embodiments, the conductive carbon-based material **104** may be configured as conductive traces or vias that may be in electrical connection with circuitry of a semiconductor die (e.g., with circuitry of an upper device region **103** of the die) to other conductive components, e.g., through openings **118** into which under bump metallization (UBM) for connection of solder bumps, or other external conductive elements (not shown) may be disposed for the component-to-component connections.

[0046] In some embodiments, the carbon-based material **104** may consist essentially of or consist of graphene with an electrical resistivity of about $10^{-6} \Omega \cdot \text{cm}$. This low electrical resistivity may be exhibited regardless of the width or thickness of the region of the carbon-based material **104**. If present, the seed material **106** may comprise, consist essentially of, or consist of nickel (Ni), a bi-layer of copper (Cu) and nickel (Ni), or another conductive material formulated for growth of the carbon-based material (e.g., the graphene) thereon.

[0047] With such low resistivity, the carbon-based material **104** may be formed to be thin (e.g., to define, with the seed material **106** (if present) a low vertical height $H_{\text{sub.R}}$). For example, the carbon-based material **104** (alone or in combination with the seed material **106**) may define a height $H_{\text{sub.R}}$ (and, therefore, the RDL **102** may define the height $H_{\text{sub.R}}$) of less than about 2000 Å (less than about 0.2 μm) (e.g., less than about 1000 Å (less than about 0.1 μm), less than about 500 Å (less than about 0.05 μm)). The height $H_{\text{sub.R}}$ may be defined entirely or primarily (e.g., at least 50%) by the carbon-based material **104**. If present, the seed material **106** may be thinner (e.g., define a lesser vertical height) than the carbon-based material **104**.

[0048] The conductive materials (e.g., the carbon-based material **104** and, if present, the seed material **106**) may be formed as discrete, electrically conductive elements **107** (e.g., conductive traces, conductive vias) in contact with a surface **109** of the semiconductor die (e.g., the surface **109** of the upper device region **103** of the semiconductor die). A lower device region **110**, which may be supported by a substrate **112**, may be between the substrate **112** and the upper device region **103**, below the RDL **102**.

[0049] Space between the conductive elements **107** may be filled with a passivation material **114**. Another passivation material **116** may overlay the passivation material **114** and the conductive elements **107** of the carbon-based material **104** and (if present) the seed material **106**, with access to the conductive elements **107** provided by the openings **118** in the other passivation material **116**. The passivation materials **114**, **116** may each or both comprise, consist essentially of, or consist of an electrically insulative material such as a nitride (e.g., silicon nitride ($\text{Si}_{\text{sub.3}}\text{N}_{\text{sub.4}}$), silicon oxynitride ($\text{SiO}_{\text{sub.x}}\text{N}_{\text{sub.y}}$), silicon carbonitride (SiCN)). In some embodiments, the passivation material **114** and the other passivation material **116** may have the same chemical composition. In other embodiments, they may have different chemical compositions.

[0050] Because the conductive material (e.g., the carbon-based material **104** and (if present) the seed material **106**) of the conductive elements **107** defines a low height $H_{\text{sub.R}}$, the passivation material **114** between the conductive elements **107** also defines the low height $H_{\text{sub.R}}$. And, because the RDL **102** comprises primarily the carbon-based material **104** with low resistivity, the other passivation material **116** above the RDL **102** materials may also be thin (e.g., define a low vertical height $H_{\text{sub.I}}$) and still exhibit sufficient electrical insulative properties. In some embodiments, the other passivation material **116** may define a height $H_{\text{sub.I}}$ of less than about 2000 Å (less than about 0.2 μm) (e.g., less than about 1000 Å (less than about 0.1 μm), less than about 500 Å (less than about 0.05 μm)).

[0051] Because both the height $H_{\text{sub.R}}$ of the conductive elements **107** (e.g., conductive traces, conductive vias) and the height $H_{\text{sub.I}}$ of the other passivation material **116** are less than 0.2 μm each, the semiconductor device package **100** may be significantly thinner than if these conductive elements **107** of the RDL **102** included a conductive material having higher electrical resistivity, such as aluminum (Al), at a thickness on the order of perhaps 3 μm to 5 μm topped with a passivation material of similar thickness. The use of the carbon-based material **104** with low

resistivity may enable a significant height form factor savings, which accommodates vertical scaling in three-dimensional stacking of semiconductor device structures.

[0052] While the upper device region **103** of FIG. **1** is illustrated without detail, FIG. **2** illustrates an enlargement. The upper device region **103** may include interdielectric materials **122**, **124**, **126** (eg., insulative materials), interleaved with and adjacent to conductive materials (eg., metals) **123**, **125**, **127** electrically connected by conductive plugs **129**.

[0053] Adjacent the upper device region **103** of the semiconductor die may be the lower device region **110**, which may include active circuitry for a semiconductor device (eg., a semiconductor memory device). For example, memory circuitry in the form of dynamic random access memory (DRAM) circuitry or NAND flash memory circuitry may be included. The active circuitry is in electrical communication with external conductive elements through electrically conductive materials (e.g., materials **123**, **125**, **127**), including the carbon-based material **104** configured as vias **105'** and conductive traces **105''** in the aforementioned portion of the RDL **102**.

[0054] With reference to FIG. **3**, another embodiment for a semiconductor device package **300** is illustrated. As with the semiconductor device package **100** of FIG. **1**, the semiconductor device package **300** may include redistribution layer **102** with discrete structures providing conductive elements **107** that include the carbon-based material **104** and, optionally, the seed material **106**. As noted above, the electrically conductive, carbon-based material **104** may be configured as vias and conductive traces, and the conductive elements **107** may be in electrical connection with other components by conductive material contacting the conductive elements **107** through openings **118** in the passivation material **114** that may extend between and overlay the discrete elements **107** to define the height $H_{sub.I}$ above the RDL **102**.

[0055] The conductive elements **107** and the passivation material **114** of the RDL **102** may be in contact with a surface **309** of a device region **310**, with the device region **310** including a stacked die structure (eg., a three-dimensional (3D) stack of semiconductor dice). An upper die **320** of the device region **310** may include a logic die while lower dice **330** of the device region **310** may include multiple DRAM dice. The die of the upper portion **320** may alternatively be referred to as a “controller” die, and the dice of the lower portion **330** may be referred to as “memory” dice when the device region **310** is characterized as having a master-slave type of architecture. Vias **340** (e.g., through-silicon-vias (TSV s)) and conductive pillars between the TSV s of the various dice may extend through the upper die **320** and the lower dice **330** of the device region **310** and contact the conductive elements **107** of carbon-based material **104** of the RDL **102**. The sides of the lower dice **330** may be bordered by a dielectric molding compound **350**, and the semiconductor device package **300** may be referred to as a “singulated cube.”

[0056] As with the semiconductor device package **100** of FIG. **1**, the semiconductor device package **300** of FIG. **3** may exhibit improved height form factor due to the thinness of the conductive elements **107** in the RDL **102** and the passivation material **114**, enabled by the use of the carbon-based material **104** with low resistivity.

[0057] While the structures of FIGS. **1** through **3** illustrate one elevation of the RDL **102** comprising the carbon-based material **104**, in other embodiments, multiple elevations within the RDL **102** may include the carbon-based material **104**, each of which may exhibit the thinness of the low height $H_{sub.R}$ and benefit from the improved height form factor due to the low resistivity of the carbon-based material **104**.

[0058] Accordingly, disclosed is a semiconductor device package comprising a redistribution layer including conductive elements comprising a conductive carbon-based material. The conductive elements of the redistribution layer define a vertical height of less than about $0.2\ \mu\text{m}$. A passivation material is above the conductive elements of the redistribution layer and defines another vertical height of less than about $0.2\ \mu\text{m}$. A device region of a semiconductor die comprising circuitry is connected to the conductive elements of the redistribution layer.

[0059] The semiconductor device packages of the disclosure (eg., semiconductor device package

100 of FIGS. **1** and **2**, semiconductor device package **300** of FIG. **3**) may be fabricated by methods that form the carbon-based material **104** with a high-temperature formation process. As used herein, the term “high temperature” means and refers to temperatures exceeding 550° C. For example, the carbon-based material **104** may be graphene formed at high temperatures. Such high temperatures may be needed to sufficiently form the graphene. Even so, by the disclosed methods, the high temperatures for forming the graphene (or other carbon-based material **104**) may not risk damage to components or materials of the device regions **103**, **110** (FIGS. **1** and **2**), **310** (FIG. **3**) because the components and materials of the device regions **103**, **110**, **310** may be formed separately from (eg., on a separate supportive wafer) or subsequently to formation of the graphene (or other carbon-based material **104**). For example, and as discussed further below, the carbon-based material **104** may be formed at a high temperature on a sacrificial substrate of an initial support wafer before, after, or while materials of the device regions **103**, **110**, **310** are formed on a device wafer. The initial support wafer and its carbon-based material **104** may be joined to the device wafer, and then sacrificial materials of the initial support wafer removed to leave the carbon-based material **104** supported on the device regions **103**, **110**, **310** of the device wafer. [0060] Accordingly, disclosed is a redistribution layer comprising at least one level of conductive elements comprising a carbon-based material. The conductive elements of a level of the at least one level define a height of less than about 0.2 μm . The redistribution layer also includes a passivation material about the conductive elements. The conductive elements are accessible through the passivation material to other conductive materials.

[0061] With reference to FIGS. **4** through **15**, illustrated are stages of a method of fabricating the semiconductor device package **100** of FIGS. **1** and **2**. With reference to FIG. **4**, a sacrificial base material **408** may be formed (e.g., by CVD) over a sacrificial substrate **410** of an initial support wafer. The sacrificial base material **408** may comprise, consist essentially of, or consist of a dielectric material (e.g., silicon dioxide (SiO_2), silicon nitride (Si_3N_4), silicon carbonitride (SiCN), silicon-carbon-oxygen (SiCO)). The sacrificial substrate **410** may comprise, consist essentially of, or consist of silicon (eg., monocrystalline silicon, polysilicon) or any another material capable of supporting the sacrificial base material **408** during subsequent processing.

[0062] The sacrificial base material **408** may exhibit a continuous surface **411** upon which the discrete structures for the conductive elements **107** and intervening passivation material **114** may be formed to fabricate an initial support wafer **500**, illustrated in FIG. **5**. As initially formed, the conductive elements **107** are inverted from their final orientation, such that they have the seed material **106** below the carbon-based material **104**.

[0063] In some embodiments, the initial support wafer **500** (FIG. **5**) may be fabricated by forming and patterning the seed material **106** and the carbon-based material **104** before forming the passivation material **114**, as illustrated in FIGS. **6** through **8**, which show additional processing stages in the portion of the initial support wafer **500** indicated by box A.

[0064] With reference to FIG. **6**, the seed material **106** may be formed (eg., by PVD, by CVD, by plating) on the surface **411** of the sacrificial base material **408**. Thus, the sacrificial base material **408** may be formulated and configured to provide a sufficient surface structure (eg., a smooth, continuous structure) upon which to form the seed material **106**. The carbon-based material **104** may be formed (eg., by microwave plasma CVD (MPCVD), by high-temperature CVD, by plasma CVD, by magnetron CVD, by plasma-enhanced CVD) on the seed material **106** to define thickness $H_{\text{sub.R}}$. The formation may be conducted at high temperatures to enable forming the carbon-based material **104** (eg., graphene), with the underlying seed material **106** used to enable the carbon-based material **104** to adhere to the structure as the material forms. In other embodiments, the sacrificial base material **408** may incorporate or be formulated to serve as a seed material for the formation of the carbon-based material **104**. So, in these embodiments, a separate seed material **106** may be omitted, and the carbon-based material **104** may be formed directly on the sacrificial base material **108**.

[0065] With reference to FIG. 7, the carbon-based material **104** and the seed material **106** may be patterned (eg., by photolithography and etching (eg., dry etching, wet etching)) to remove select portions of the materials **104**, **106** and define the discrete structures for the conductive elements **107** and openings **702** in which select areas of the surface **411** of the sacrificial base material **408** are exposed. For example, in embodiments in which the carbon-based material **104** is graphene, it may be patterned by plasma dry etching, and then underlying portions of the seed material **106** may be removed by chlorine-based etching. In some embodiments, some portions of the underlying sacrificial base material **408** may also be removed to form the openings **702**. The openings **702** may define widths of less than one micrometer (less than 1 μm), and the thinness of the carbon-based material **104** and the seed material **106** may accommodate forming the openings **702** without difficulties normally associated with etching through thicker materials (eg., complex masking materials, sloping sidewalls, collapsing structures).

[0066] With reference to FIG. 8, the openings **702** (FIG. 7) may then be filled by forming (eg., by CVD) the passivation material **114** therein and, optionally, above the conductive elements **107**, as shown in structure **800** illustrated in FIG. 8. In embodiments in which the passivation material **114** is formed to overlay the conductive elements **107**, the structure **800** may then be subjected to planarization (e.g., CMP) to expose the conductive elements **107** through the passivation material **114**, as in the structure of the initial support wafer **500** of FIG. 5.

[0067] In other embodiments, the structure of the initial support wafer **500** (FIG. 5) may be formed by first forming and patterning the passivation material **114** and then forming the discrete structures for the conductive elements **107** of the carbon-based material **104** and the seed material **106** therein, as illustrated in FIGS. 9 through 11, which show additional processing stages in the portion of the initial support wafer **500** indicated by box A. For example, the passivation material **114** may be formed to a thickness of $H_{\text{sub.R}}$, as illustrated in FIG. 9; patterned to form openings **1002** exposing areas of the surface **411** of the sacrificial base material **408**, as illustrated in FIG. 10; and then, as illustrated in FIG. 11, the seed material **106** and the carbon-based material **104** may be sequentially formed in the openings **1002** (FIG. 10) to form the discrete structures for the conductive elements **107** exhibiting the low thickness $H_{\text{sub.R}}$ and resulting in the initial support wafer **500** of FIG. 5.

[0068] The initial support wafer **500** (whether formed by the stages illustrated in FIGS. 6 through 8, by the stages illustrated in FIGS. 9 through 11, or by some other method) may then be flipped (e.g., inverted), as indicated by arrow B in FIG. 12, and joined to a device wafer **1200** that has been separately fabricated to include the upper device region **103** and the lower device region **110** supported by the substrate **112**. More particularly, a surface **501** (FIG. 12) of the initial support wafer **500** (eg., what was an upper surface) may be brought in contact with an upper surface **1201** of the device wafer **1200**, and the two surfaces **501**, **1201** bonded together to form joined structure **1300** of FIG. 13. In some embodiments, the upper surface **1201** of the device wafer **1200** may include dielectric material (e.g., interdielectric material **122** (FIG. 2)) configured or formulated to have dangling bonds that will, when brought in contact with the materials of the surface **501** of the initial support wafer **500** (eg., the passivation material **114**) chemically bond thereto to affix the initial support wafer **500** to the device wafer **1200**. As such, the initial support wafer **500** may be bonded to the device wafer **1200** by “dielectric bonding.”

[0069] In the joined structure **1300**, the conductive elements **107** of the RDL **102** include the carbon-based material **104** on (eg., directly on) the upper surface **1201** of the device wafer **1200**, and the seed material **106** above the carbon-based material **104**. At this stage, the sacrificial base material **408** overlays the RDL **102** and the passivation material **114**, and the sacrificial substrate **410** overlays the sacrificial base material **408**.

[0070] Because the initial support wafer **500** and its carbon-based material **104** are fabricated separately from the fabrication of the device wafer **1200**, the high-temperature formation processes used to form the carbon-based material **104** do not risk detrimentally impacting the materials or

components of the device region (e.g., the materials and components of the upper and lower device regions **103**, **110**). Therefore, even graphene or another carbon-based material **104** that may require high temperatures for formation, can be used in the RDL **102** without the high-temperature formation processes damaging other materials of the semiconductor device being fabricated.

[0071] The sacrificial materials of the initial support wafer **500** may then be removed to expose the conductive elements **107** of the RDL **102** and the intervening passivation material **114**. For example, the sacrificial substrate **410** may be removed (e.g., by etching (e.g., wet etching, dry etching), by grinding) to form structure **1400** of FIG. **14**; and, the sacrificial base material **408** may then be removed (e.g., by etching (e.g., wet etching, dry etching), by grinding) to form structure **1500** of FIG. **15**. In some embodiments, removal of the sacrificial substrate **410** and removal of the sacrificial base material **408** may be two distinct stages. In other embodiments, both the sacrificial substrate **410** and the sacrificial base material **408** may be removed together or substantially together to form structure **1500**.

[0072] The other passivation material **116** (FIG. **1**) may then be formed on the conductive elements **107** (e.g., on or directly on the seed material **106** of the RDL **102**) and the passivation material **114** and then patterned to form the openings **118** (FIG. **1**), to form the semiconductor device package **100** of FIG. **1**.

[0073] Accordingly, disclosed is a method for fabricating a semiconductor device package. The method comprises forming a carbon-based material on a sacrificial substrate. The method also includes inverting the sacrificial substrate and bonding the sacrificial substrate to another wafer comprising semiconductor device components. After the bonding, the sacrificial substrate is removed to leave the carbon-based material over the other wafer.

[0074] In some embodiments, the conductive elements **107** with intervening passivation material **114** may not be formed until after an initial support wafer is flipped and joined to the device wafer **1200**. For example, with reference to FIG. **16**, an initial support wafer **1600** may be fabricated by forming the seed material **106** over the sacrificial base material **408** and forming the carbon-based material **104** over the seed material **106**. An upper surface **1601** of the initial support wafer **1600** may, therefore, be a continuous surface of the carbon-based material **104** or, in some embodiments, a dielectric material (not illustrated) formed on the carbon-based material **104**. The initial support wafer **1600** may then be flipped (as indicated by arrow B) and joined to the upper surface **1201** of the device wafer **1200**, forming joined structure **1700** of FIG. **17**. The sacrificial substrate **410** and the sacrificial base material **408** may then be removed, as described above, to expose the seed material **106** above the carbon-based material **104**, as illustrated in FIG. **18**.

[0075] The seed material **106** and the carbon-based material **104** may then be patterned, in the same manner described above with respect to FIG. **7**, by forming openings **1902** (FIG. **19**) that expose areas of the upper surface **1201** of the device wafer **1200**. The passivation material **114** may then be formed to fill the openings **1902** and overlay the conductive elements **107** of the RDL **102** and patterned to form the openings **118**, to form semiconductor device package **2000** of FIG. **20**. The semiconductor device package **2000** of FIG. **20** may be the same as that the semiconductor device package **100** of FIG. **1**, except—in the semiconductor device package **2000** of FIG. **20**—the passivation material **114** may be continuous both between and above the conductive elements **107**, instead of—as in the semiconductor device package **100** of FIG. **1**—the passivation material **114** between the conductive elements **107** being separately formed from the other passivation material **116** above the conductive elements **107**. Nonetheless, in embodiments in which the passivation material **114** and the other passivation material **116** have the same composition, there may be no visual distinction between the semiconductor device packages **100**, **2000**.

[0076] In some embodiments, some or all of the seed material **106** may be removed with or subsequent to removal of the sacrificial materials (e.g., the sacrificial substrate **410** (FIG. **13**) and the sacrificial base material **408** (FIG. **14**)) of the initial support wafer **500**. For example, further to the method illustrated in FIGS. **4** through **15** and with returned reference to FIG. **15**, the exposed

seed material **106** and adjacent portions of the passivation material **114** may be removed (e.g., by CMP, by grinding, by etching) to leave only the carbon-based material **104** exposed, as illustrated in FIG. **21**. Thus, conductive elements **2107** of redistribution layer **2102** may consist of the carbon-based material **104** and may define a height $H_{sub.R'}$. Height $H_{sub.R'}$ may be thin, at less than about 2000 Å (less than about 0.2 μm) (e.g., less than about 1000 Å (less than about 0.1 μm), less than about 500 Å (less than about 0.05 μm)). The other passivation material **116** may then be formed on (e.g., directly on) the carbon-based material **104** and the passivation material **114** to the thickness of $H_{sub.I}$ and patterned to form openings **118**, as illustrated in FIG. **22**, forming semiconductor device package **2200** with the significant height form factor savings.

[0077] As another example, further to the method illustrated in FIGS. **4**, **6**, and **16** through **20**, and with returned reference to FIG. **18**, the seed material **106** may be removed (e.g., by CMP, by grinding, by etching) to leave the carbon-based material **104** exposed and defining the height $H_{sub.R'}$, as illustrated in FIG. **23**. The carbon-based material **104** may then be patterned (e.g., by photolithography and etching (e.g., dry etching, wet etching)) to form openings **2402** exposing areas of the upper surface **1201** of the device wafer **1200**, as illustrated in FIG. **24**. The passivation material **114** may then be formed to fill the openings **2402** and overlay the conductive elements **2107** of the RDL **2102** consisting of the carbon-based material **104**. The passivation material **114** may then be patterned (e.g., by photolithography and etching) to form the openings **118**. The resulting semiconductor device package **2500** (FIG. **25**) may also benefit from the significant height form factor savings enabled by using the carbon-based material **104** with low resistivity and low height.

[0078] In some embodiments, the other passivation material **116**, which defines the height $H_{sub.I}$ above the conductive elements **107**, may be formed prior to forming the conductive materials of the conductive elements **107** and prior to the flip and bonding. For example, with reference to FIG. **26**, the other passivation material **116** may be on (e.g., directly on) the sacrificial base material **408**, such as to form a continuous film defining the height $H_{sub.I}$. The seed material **106** and the carbon-based material **104** may be formed, sequentially, on the other passivation material **116**, as illustrated in FIG. **27**. The seed material **106** and the carbon-based material **104** may be patterned to form openings **2802** exposing areas of the other passivation material **116**, as illustrated in FIG. **28**. The passivation material **114** may then be formed to at least fill the openings **2802**, forming initial support wafer **2900**, as illustrated in FIG. **29**, with an upper surface **2901** exposing the carbon-based material **104** and the passivation material **114** between the conductive elements **107**. The initial support wafer **2900** may then be inverted and joined to the device wafer **1200** with the upper surface **2901** of the initial support wafer **2900** contacting the upper surface **1201** of the device wafer **1200**, as illustrated in FIG. **30**. The sacrificial materials of the initial support wafer **2900** (e.g., the sacrificial substrate **410**, the sacrificial base material **408**) may then be removed, leaving the other passivation material **116** over the conductive elements **107** that include the carbon-based material **104**, as illustrated in FIG. **31**. The passivation material **116** may then be patterned to form the openings **118**. The resulting structure may be equivalent or identical to the semiconductor device package **100** of FIG. **1**.

[0079] As another example, with reference to FIG. **32**, the passivation material **114** may be formed over the sacrificial base material **408** to a height of $H_{sub.R} + H_{sub.I}$. Portions of the passivation material **114** may be removed (e.g., etched) to form openings **3302** defined wholly in the passivation material **114**, as illustrated in FIG. **33**. The seed material **106** and the carbon-based material **104** may be formed in the openings **3302** to form the conductive elements **107** of the RDL **102**, as illustrated in FIG. **34**, forming initial support wafer **3400** with a continuous region of the passivation material **114**. The initial support wafer **3400** may then be inverted and joined to bond an upper surface **3401** of the initial support wafer **3400** to the upper surface **1201** of the device wafer **1200**, as illustrated in FIG. **35**. The sacrificial materials (e.g., the sacrificial substrate **410** and the sacrificial base material **408**) may then be removed, leaving the passivation material **114** above

and between the conductive elements **107**, as illustrated in FIG. **36**. The passivation material **114** may then be patterned to form the openings **118**. The resulting structure may be equivalent or identical to the semiconductor device package **2000** of FIG. **20**.

[0080] The carbon-based material **104** in the RDL **102** may also be used in chip-on-wafer embodiments in which the components of a device layer (e.g., device region **310** (FIG. **3**)) are constructed above the initial support wafer after forming the carbon-based material **104** but before flipping the wafer. With reference to FIG. **37**, an initial support wafer **3700** may be fabricated, eg., using the method illustrated by FIGS. **26** through **29** or using the method illustrated by FIGS. **32** through **34**. The materials and components of the device region **310** (FIG. **3**) may then be formed (eg., by methods such as those described in U.S. patent application Ser. No. 15/976,398, filed May 10, 2018) over the initial support wafer **3700**.

[0081] For example, the materials and components of the upper portion **320** of the device region **310** (FIG. **3**) may be formed on or joined to an upper surface **3701** of the initial support wafer **3700**, as illustrated in FIG. **38**. In some embodiments, the upper portion **320** may be formed on another substrate **3802** (e.g., another silicon wafer), separate from the initial support wafer **3700**, with such other substrate **3802** being processed to expose a portion of the via **340** (FIG. **3**) through a backside dielectric material **3814** before the structure is bonded to the initial support wafer **3700**. The resulting structure **3800** includes multiple die locations **3810**, each with an active surface comprising active circuitry **3806** (e.g., memory circuitry in the form of DRAM circuitry or NAND flash memory circuitry). The active circuitry **3806** may later be in electrical communication with external conductive elements in the form of, eg., copper pillars that may occupy the openings **118** (FIG. **3**) communicating to the conductive elements **107**. Between the die locations **3810** are scribe areas **3820**, delineating so-called “streets.”

[0082] On the structure **3800** (eg., on the backside dielectric material **3814**) may be formed materials and components of another portion **3930** that will become part of the lower dice **330** (FIG. **3**) of the semiconductor device package **300** (FIG. **3**). The vias **340** may be formed to extend from the carbon-based material **104** in the initial support wafer **3700** to an upper die of the other portion **3930**, as illustrated in FIG. **39**. Openings **3902** may be formed in the scribe areas **3820**, extending through the other portion **3930** to the upper portion **320**. The openings **3902** may be filled with the molding compound **350**, as illustrated in FIG. **40**. The structure may then be inverted and the sacrificial materials (the sacrificial substrate **410** and the sacrificial base material **408**) removed, as illustrated in FIG. **41**. The openings **118** may be formed through the passivation material **114**, and the structure may then be separated, or “singulated” at cut line C through scribe areas **3820** (FIG. **40**) into individual semiconductor dice (eg., individual, singulated semiconductor device packages **300**). The singulation may be effected with a very narrow dicing saw blade width.

[0083] Notably, by the method of FIGS. **37** through **41**, the high-temperature processes used to form the carbon-based material **104** do not risk detriment to the components or materials of the device region **310** because the device region **310** is formed either subsequent to forming the carbon-based material **104** or are formed separately and subsequently bonded above the already-formed carbon-based material **104**.

[0084] It should be recognized that the initial support wafer **3700** portion of the structure of FIG. **40** may be otherwise formed by any of the other aforementioned methods for forming an initial support wafer. For example, initial support wafer **3700** may be first formed as the initial support wafer **500** of FIG. **5** and the upper portion of the passivation material **114** (i.e., the portion defining height H.sub.I) may be formed after the sacrificial substrate **410** and the sacrificial base material **408** are removed. In addition, in some embodiments, the seed material **106** may be removed before forming such additional portion of the passivation material **114**, as in the methods discussed above with respect to FIGS. **21** through **25**.

[0085] Accordingly, disclosed is a method for fabricating a semiconductor device package. The method comprises forming an initial support wafer. Forming the initial support wafer comprises

forming a conductive carbon-based material on a sacrificial substrate. A surface of the initial support wafer is bonded to another wafer comprising semiconductor device components. The sacrificial substrate is removed, and a passivation material is formed over the conductive carbon-based material.

[0086] Semiconductor devices (eg., the semiconductor device packages **100** (FIG. 1), **300** (FIG. 3), **2000** (FIG. 20), **2200** (FIG. 22), **2500** (FIG. 25)), in accordance with embodiments of the disclosure, may be used in embodiments of electronic systems of the disclosure. For example, with reference to FIG. 42, shown is a block diagram of an illustrative electronic system **4200**, according to embodiments of the disclosure. The electronic system **4200** may comprise, for example, a computer or computer hardware component, a server or other networking hardware component, a cellular telephone, a digital camera, a personal digital assistant (PDA), a portable media (eg., music) player, a Wi-Fi- or cellular-enabled tablet (eg., an IPAD® or SURFACE® tablet, an electronic book, an electronic reader, a navigation device). The electronic system **4200** includes at least one memory device **4202**. The memory device **4202** may include, for example, an embodiment of a semiconductor device package (eg., the semiconductor device packages **100** (FIG. 1), **300** (FIG. 3), **2000** (FIG. 20), **2200** (FIG. 22), **2500** (FIG. 25)) described herein. Such memory device may, optionally, include dice configured for other functions (eg., a logic die, an RF die, or both). The electronic system **4200** may further include at least one electronic signal processor device **4204** (often referred to as a “microprocessor”). The electronic system **4200** may further include one or more input devices **4206** for inputting information into the electronic system **4200** by a user, e.g., using a mouse or other pointing device, a keyboard, a touchpad, a button, or a control panel. The electronic system **4200** may further include one or more output devices **4208** for outputting information (e.g., visual or audio output) to a user by, eg., a monitor, a display, a printer, an audio output jack, a speaker, etc. In some embodiments, the input device **4206** and the output device **4208** may comprise a single touchscreen device that can be used both to input information to the electronic system **4200** and to output visual information to a user. The input device **4206** and the output device **4208** may communicate electrically with one or more of the memory devices **4202** and the electronic signal processor device **4204**. It is also contemplated that, in lieu of separate memory and signal processor devices **4202** and **4204**, a single assembly may be configured as a system (e.g., the electronic system **4200**) in a package including a processor and/or other die functionalities.

[0087] Accordingly, disclosed is an electronic system comprising at least one input device, at least one output device, and a processor device operably coupled to the at least one input device and to the at least one output device. The electronic system also comprises a semiconductor device that comprises at least one memory die, a redistribution layer, and a passivation material. The redistribution layer comprises conductive elements of a conductive carbon-based material. The conductive elements define a height of less than about 0.2 μm and are electrically connected to the at least one memory die. The passivation material electrically isolates the conductive elements of the redistribution layer and also defines a height of less than about 0.2 μm .

[0088] While certain illustrative embodiments have been described in connection with the figures, those of ordinary skill in the art will recognize and appreciate that embodiments encompassed by the disclosure are not limited to those embodiments explicitly shown and described herein. Rather, many additions, deletions, and modifications to the embodiments described herein may be made without departing from the scope of embodiments encompassed by the disclosure, such as those hereinafter claimed, including legal equivalents. In addition, features from one disclosed embodiment may be combined with or substituted for features of another disclosed embodiment while still being encompassed within the scope of the disclosure.

Claims

- 1.** A redistribution layer, comprising: a single level of conductive elements horizontally distributed within a passivation material, the conductive elements individually comprising a conductive region comprising a carbon-based material, the conductive region defining a vertical height of less than about 0.2 μm from a lower surface of the conductive region to an upper surface of the conductive region, the passivation material continuing from between the conductive elements to adjacent at least one of the lower surface and the upper surface of the conductive region.
- 2.** The redistribution layer of claim 1, wherein the conductive regions individually comprise a seed material on the carbon-based material.
- 3.** The redistribution layer of claim 2, wherein the seed material is vertically between the carbon-based material and at least a portion of the passivation material.
- 4.** The redistribution layer of claim 1, wherein the passivation material defines a vertical height of less than about 0.2 μm between an upper surface of the passivation material and the at least one of the lower surface and the upper surface of the conductive region.
- 5.** The redistribution layer of claim 1, wherein the carbon-based material consists essentially of graphene.
- 6.** The redistribution layer of claim 1, wherein the conductive elements are substantially free of aluminum (Al).
- 7.** The redistribution layer of claim 1, wherein the conductive elements consist essentially of the carbon-based material.
- 8.** A microelectronic device package, comprising: a redistribution layer comprising an array of conductive elements arranged in a single level and distributed within a passivation material, the conductive elements individually comprising a conductive carbon-based material defining a vertical height of less than about 0.2 μm between a first surface of the conductive element to a second surface of the conductive element; and a microelectronic device region in physical contact with the first surface of the conductive elements of the redistribution layer, the passivation material extending over a portion of the second surface of the conductive elements.
- 9.** The microelectronic device package of claim 8, wherein the conductive elements individually comprise a seed material on the conductive carbon-based material.
- 10.** The microelectronic device package of claim 9, wherein the seed material and the conductive carbon-based material have substantially coplanar sidewalls.
- 11.** The microelectronic device package of claim 9, wherein the seed material comprises at least one material chosen from nickel (Ni) and copper (Cu).
- 12.** The microelectronic device package of claim 9, wherein the seed material comprises nickel (Ni) and copper (Cu).
- 13.** The microelectronic device package of claim 8, wherein the conductive carbon-based material comprises graphene.
- 14.** The microelectronic device package of claim 8, wherein the passivation material defines openings extending therethrough to the second surface of the conductive elements.
- 15.** A method of forming microelectronic device package comprising a redistribution layer on a device region of a structure, the method comprising: forming an array of conductive elements arranged in a single level on a surface of the structure and horizontally distributed in a passivation material, at least some of the conductive elements being in direct physical connection with circuitry of the device region of the structure, the conductive elements comprising a conductive carbon-based material, the passivation material extending from between the conductive elements of the array of conductive elements to directly vertically adjacent at least a portion of a first surface of the conductive elements, the conductive elements defining a vertical height of less than about 0.2 μm from the first surface of the conductive elements to a second surface of the conductive elements.
- 16.** The method of claim 15, wherein forming the array of conductive elements arranged in the single level on the surface of the structure and horizontally distributed in the passivation material

comprises: forming at least one conductive material on a sacrificial structure to form an initial support wafer; bonding a surface of the initial support wafer to the surface of the structure; removing the sacrificial structure; patterning the at least one conductive material to form the array of conductive elements arranged in the single level on the surface of the structure; and forming the passivation material to be horizontally between the conductive elements of the array of conductive elements to be directly vertically adjacent the at least the portion of the first surface of the conductive elements.

17. The method of claim 16, wherein forming the at least one conductive material on the sacrificial structure comprises: forming a seed material on a surface of the sacrificial structure; and forming a conductive carbon-based material on the seed material.

18. The method of claim 16, further comprising, before patterning the at least one conductive material, removing the seed material, wherein patterning the at least one conductive material comprises patterning the conductive carbon-based material to form the array of conductive elements.

19. The method of claim 15, wherein forming the array of conductive elements arranged in the single level on the surface of the structure and horizontally distributed in the passivation material comprises: forming an initial support wafer, comprising: forming at least one conductive material on a sacrificial structure; patterning the at least one conductive material to form the array of conductive elements arranged in the single level on the surface of the sacrificial structure; and forming the passivation material horizontally between the conductive elements of the array of conductive elements; bonding a surface of the initial support wafer to the surface of the structure; and removing the sacrificial structure.

20. The method of claim 19, further comprising, after removing the sacrificial structure, forming an additional amount of the passivation material on the passivation material already formed and directly vertically adjacent the at least the portion of the first surface of the conductive elements.
